

AVIRAL NIGAM

AI SAFETY RESEARCH

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My focus is AI safety. I study reasoning faithfulness and confidence calibration through behavioral explainability and LLM evaluation, and use red teaming to uncover model failures. I'm also interested in how RLHF shapes behavior and in instrumental convergence: why systems with different goals may develop similar incentives to seek power or preserve themselves.

EDUCATION

Manipal Institute of Technology, MAHE

B.Tech in Data Science and Engineering | Expected July 2028

Coursework: Deep Learning, Statistical Modeling, Linear Algebra & Probability, Data Structures & Algorithms.

Shortlisted, Algoverse AI Research Fellowship, Fall 2026 cohort.

SELECTED AI SAFETY & EVALUATION PROJECTS

Missing-Information Evaluation | Qwen3-4B-Instruct-2507

github.com/Aviral0gHub/fit-missing-information-eval

- Built 20 policy situations with complete, underspecified, and misleading variants (60 prompts); compared an explicit-omission control with a follow-up using silent omissions and neutral instructions.
- Complete-case accuracy stayed at 20/20. In the follow-up, 22/40 incomplete cases received unsupported definite answers; 21 had confidence of at least 80%. Saved raw outputs and reproducible analysis.

Scope: one model and a small synthetic dataset. Both omission cues and instructions changed between rounds; their individual effects are not isolated.

FactTrace | Claim-level factuality verification

- Built a pipeline for claim decomposition, FAISS evidence retrieval, and NLI classification; includes a TF-IDF/heuristic baseline and dense retrieval with RoBERTa-large-MNLI.
- Matched author-assigned step labels on 57/65 extracted claims (87.7%). Traces contain deliberately placed errors; this verifies factuality, not internal reasoning faithfulness.

github.com/Aviral0gHub/facttrace

CalibCheck | Confidence calibration

- Evaluated Qwen2.5-0.5B-Instruct on 60 questions across six domains using ECE, Brier score, and reliability diagrams. Verbalized-confidence ECE was 0.1432; mean confidence exceeded accuracy by 14.3 points.
- Sports showed 30% accuracy versus 87.5% confidence on ten questions. Documented small-sample limits and the cross-model confound in the raw-probability comparison.

github.com/Aviral0gHub/calibcheck

PUBLICATION & DATASET CONTRIBUTION

Last Translation Benchmark | Dataset co-author | September 2026

V. Zouhar et al., Last Translation Benchmark, arXiv:2609.04173 [cs.CL], 2026. Named on the dataset-author list, ordered by contribution.

- Contributed adversarial text examples to a multilingual, multimodal translation benchmark with human-authored inputs and handcrafted verification rules.

arxiv.org/abs/2609.04173 | Dataset | Leaderboard

TECHNICAL SKILLS

Evaluation: behavioral tests, adversarial examples, calibration, claim verification, NLI.

ML: PyTorch, Hugging Face Transformers, Sentence-Transformers, FAISS, scikit-learn.

Languages & tools: Python, Java, Bash, Linux, Git, CUDA.